

Geomagnetic and Atmospheric Electrical Parameter Variations observed at Maitri, Antarctica

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ABSTRACT

Surface measurements of the atmospheric electrical parameters, viz. ionosphere to ground currents, electric field and geomagnetic field were carried out at the Indian station, Maitri (Geog. 70.75° S, 11.75° E, 117 m above mean sea level), Antarctica. The averaged pattern of diurnal variations in the electric field and Maxwell current density averaged over selected fair weather days nearly follows the Carnegie curve, with a minimum around 0300 UT and a maximum around 1900 UT. However during magnetically disturbed days, there is a departure from universal diurnal variation in both the electric field and vertical conduction current, which is attributed to the influence of solar wind-magnetosphere-ionosphere coupling. Examining the cause in detail will require further investigation. The correlation coefficient between the current and electric field had a high value (more than 0.75) for all these days. The measured atmospheric electric field and geomagnetic field variations were compared. The experimental data used for the analysis were of 2004 and 2005. The mean daily variation of atmospheric electric field and the horizontal magnetic components showed a coincidence for the selected fair weather days during magnetic quiet days. Some examples are presented in this paper. Further measurements and studies involving statistical analysis are needed. This study will help us to examine the impact of upper atmospheric contribution to the geo-electrical parameters during different geomagnetic conditions.

INTRODUCTION

Wilson (1925) suggested that, the worldwide thunderstorm activity maintains the global electric potential between the ionosphere and the Earth's surface. Tropical thunder- storms are considered as a main source of electric field in the lower atmosphere, drawing current upward to the ionosphere. In this way, the global thunderstorm activity is able to maintain a time varying electric potential difference of ~300 kV, directed downward between the

equipotent surfaces of the ionosphere and the ground in fair-weather regions (Adlerman et al., 1996). The atmospheric vertical electric field, conductivity and total current density are the three closely related parameters of the atmospheric electricity that are required to be observed for obtaining an adequate description of the fair-weather atmospheric electric circuit (Israel, 1973).

The Global Electric Circuit (GEC) links the lower atmosphere with the ionosphere and the magnetosphere. Measurement of atmospheric electrical parameters is handy for investigating the linkage between these regions. Long-term measurements would be considered useful for addressing some of the problems associated with global change. The measuring site needs to be free of atmospheric aerosols and convective activity, because they obscure the weak signatures representing the global thunderstorm activity (Israel, 1973). Measurements from Antarctica will allow us to investigate global and local components of global electric circuit. The local components, viz. blizzard, wind turbulence, snowfall and clouds, generate a significant effect on electrical processes and enable us to understand the electrical climate of the Antarctic plateau (Byrne et al., 1993).

There is almost no thunderstorm activity at high latitude. Therefore atmospheric currents in these regions can be regarded as a closure segment of the global electric circuit. The global circuit involves lower atmosphere generators and upper atmosphere generators, the latter being significant in the polar cap region. The atmospheric electrical measurements made in Antarctica are useful in investigating large scale electrical processes unique to high altitude (Byrne et al., 1991). The down ward air-earth current being delivered to the surface of the Antarctica plateau is larger than the global average owing to its high latitude. Because of its orography, Antarctica is strongly coupled to the GEC. Secondly, Antarctic plateau supports a desert like climate with clear skies and very low atmospheric aerosol content. In summer, the prevailing winds are light and flowing in a nearly constant direction and the atmosphere is relatively free of turbulent and convective motions (Byrne et al., 1993). Thus the wind induced conductivity fluctuations that contribute to meteorological noise in the measured atmospheric electric parameters are minimized when experiments are conducted in Antarctic. In this paper we report observations of the atmospheric electric field and magnetic field made at the Indian Antarctic research station Maitri during the year 2004 and 2005. Maitri is located on Schirmacher oasis and is a rocky area with little snow formation during the year.

DESCRIPTION OF THE INSTRUMENTS AND SITE

Maxwell current in the air is the most informative parameter of the global atmospheric electric circuit (Ruhnke, 1969). In global atmospheric measurements, an important quality of a method is the capacity to suppress disturbances of local origin. Ground-based measurements are widely used because they are inexpensive and can provide continuous long-term recordings. The most important ground-based sensors are Wilson plate (Israel, 1973), the horizontal long-wire antenna (Kasemir, 1955; Ruhnke, 1969) and spherical shell in the form of two hollow hemispheres. The long-wire antenna is considered as a good alternative since it allows for the suppression of local disturbances by averaging the vertical current over a large area (Ruhnke, 1969). The horizontal long wire antenna, if placed in the atmosphere, will closely follow the electrical current variations of the atmosphere after the initial net charge on the antenna leaks off. When the antenna is shorted to the ground through a resistor it will pick up a certain amount of current proportional to the air-earth current density. In this experiment we are using horizontal wire antenna of 41.5 m in length and 3 mm in diameter, to collect incoming charges from the atmosphere. The sensor is supported 2 m above the ground by means of masts that are electrically separated by Teflon rods. It is connected to the electrometer (Model AD 549), that has high input impedance of the order of $10^9 \Omega$ and permits extremely low input bias current (10^{-14} A). The electrometer circuit is fabricated on a Teflon sheet which has high resistance to avoid the leakage current. The electrometer converts the current into voltage and is capable of measuring the current from pA to few nA with the high feedback resistance (5 G Ω). The electrometer output signal is connected to a unity gain operational amplifier (LM 308) and is to take filtered by the low pass filter, with a cut-off frequency of 5Hz (3dB) at the input of ADC, which is 100 m away from the preamplifier. The filtered signal is fed to 12 bit ADC (AD574) with 2.44mV resolution which is mounted in the personal computer. The computer records the signal at a sampling interval of one second. (Panneerselvam et al., 2003).

The atmospheric electric field has been measured with a vertical ac field mill made of nonmagnetic stainless steel to reduce the contact potential (Willett and Bailey, 1983). The sensor plates and rotor of the field mill are of 8.5 cm diameter. The shaft of the ac motor (220V, 50Hz, 1500 rpm) is grounded with a carbon brush. The signal from the field mill is amplified with Op-Amp 1596 which is mounted inside the field mill. Further details are available in Deshpande et al., (2001) The Indian Antarctic station, Maitri

is located in the Schirmacher Oasis in the Dronning Maud Land, East Antarctica (117 m above the mean sea level). Antarctica has only around 2% of its area that is free of ice. The nearest steep cliff of the east-west trending glacier on the southern side of the station is more than 700 m away from the station and is 300 m in height. The snow-covered surface during summer season was more than half a kilometer away from the station. The instruments were installed on barren land near to the station. The surface of the station area is mainly covered by sandy and loamy sand types of soil. The solar zenith angle at Maitri varies from 48° to 88° during summer months. There is no sunset till the third week of January, but periods of short nights slowly increased during February. The variations in surface meteorological parameters like wind speed, wind direction, snowfall, pressure, visibility and all other parameters are systematically monitored by the India Meteorological Department. The cloud cover over the station occurs mainly under the influence of subpolar low-pressure systems and shows an alternating sequence of the sky changing over to the overcast and then clearing as the system moves away (Deshpande and Kamra, 2001).

RESULTS AND DISCUSSION

The selection of the fairweather days has been made with the standard procedure adopted in the past (Reiter 1985). In the present study days during which both fairweather and magnetic quiet conditions day prevailed all through the day were considered, and the hourly averages were computed to yield the diurnal variation. The International quiet and disturbed days are downloaded from the website “World Data Center for Geomagnetism, Kyoto”. The selection of the quietest days (Q-days) and most disturbed days (D-days) of each month is deduced from the K_p indices. The global K_p index is obtained as the mean value of the disturbance levels in the two horizontal field components, observed at 13 selected stations. The quiet and disturbed days are calculated as follows:

- i) The sum of the eight K_p values
- ii) The sum of squares of the eight K_p values
- iii) The maximum of the eight K_p values.

According to each of these criteria, a relative *order number* is assigned to each day of the month, the three order numbers are averaged and the days

with the lowest and the highest mean order numbers are selected as the five quietest and the five most disturbed days.

The selection of the fair-weather days has been made with the standard procedure adopted in the past (Reiter, 1985). We define a fair-weather day when there is no snowfall at the measuring site, high clouds less than 3 octas throughout the day, and wind speed less than or equal to 10 ms^{-1} . In this present study fairweather conditions prevailed all through the day and the hourly averages were computed to yield the diurnal variation. The fair-weather days were further classified into geomagnetic disturbed condition and geomagnetic quiet condition with respect to solar activity. Features corresponding to the UT variation were then looked for. Mean diurnal variation of atmospheric electric field during magnetic quiet and disturbed days is shown in Fig. 1. The atmospheric electric field has a minimum at about 0300 UT and a maximum about 2100 UT. This diurnal trend is consistent with the familiar “Carnegie curve” variation. This variation

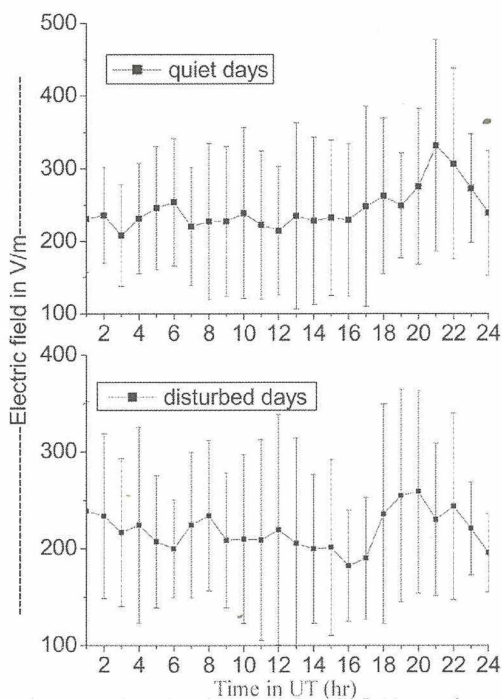


Fig. 1: Diurnal variation of electric field on quite and disturbed days for selected fair weather days during 2004 at Maitri, Antarctica.

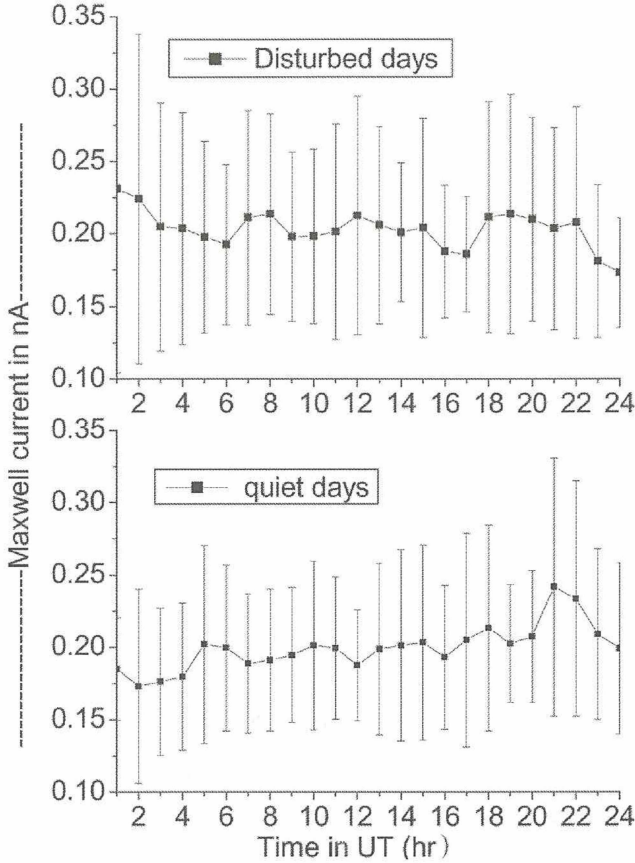
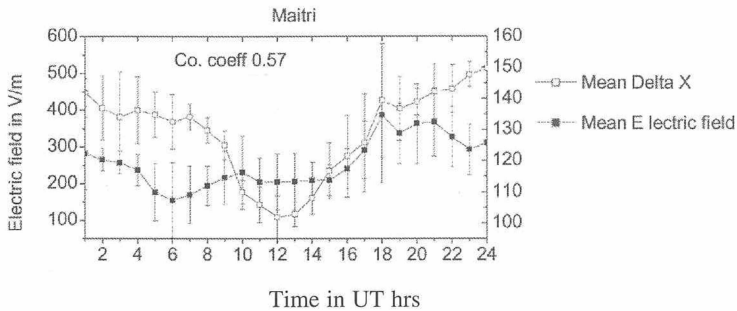


Fig. 2: Diurnal variation of atmospheric maxwell current of quite and disturbed days for selected fair weather days at Maitri, Antarctica.

has been widely observed, and according to classical theory, is generally attributed to the variation with time of day of the number of thunderstorms across the globe. If thunderstorm activity were responsible for the generation of global electric circuit and its variation with time, one would expect a maximum in the measured electrical parameters near 2100 UT and a minimum near 0300 UT. The diurnal variation of Maxwell current is shown in Fig. 2. This study helps to find out the magnetospheric contribution to the GEC. We have tried to compare some results of geomagnetic and atmospheric electric field measurements obtained at Maitri during January 2005. The results are shown in Fig. 3 for the selected fair-weather days during magnetic quiet conditions. The mean variation of both parameters



Magnetic quiet days: 24 to 27 January 2005 February 12, 2005

Fig. 3: Diurnal variation of magnetic field and electric field

shows similar variation and the correlation coefficient between these measured parameters has a high value (more than 0.5) for the selected fair-weather days. The electric field against the time derivative of the horizontal magnetic field variation is shown in Fig. 4. The preliminary results obtained in Maitri need to be examined further with statistical treatment of the obtained results, comprising spectral analyses and properties of polarization of the geomagnetic field and its influence on atmospheric electric field. If these investigations yield more corroborated data on the behavior of geomagnetic disturbances in the atmosphere below the ionosphere, we may expect to get additional information. The expected information would concern the effective ionospheric boundary conditions for the magnetospheric disturbances and may also be useful for a better understanding of physical nature of geomagnetic pulsations.

CONCLUSION

With the selected fair-weather days data we have made an attempt to explain the influence of upper atmospheric contribution during geomagnetically disturbed periods. During geomagnetically quiet days the variations of measured atmospheric electrical parameters are similar to the famous fair weather “Carnegie curve”. But, during geomagnetically disturbed days the diurnal pattern is different from the Carnegie curve and is modified by the ionospheric / magnetospheric contributions. With the continuous measurements of atmospheric electrical parameters and geomagnetic field variations, there is scope for addressing to the problems related to the modulation of GEC by the influence of magnetosphere- ionosphere coupling processes on the near surface electrical parameters at Antarctica.

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REFERENCES

1. Adlerman, E. J. and E. R. Williams, Seasonal variation of the global electric circuit, *J. Geophys. Res.* 101, 29,679-29,688, 1996.
2. Byrne, G.J., Benbrook, J.R., Bering, E.A., Balloon observations of atmospheric electricity above the South pole: vertical electric field, conductivity and conduction current, *J. Atmos. Terr. Phys.*, 53(9), 859-868, 1991.
3. Byrne, G. J., J. R. Benbrook, E. A. Bering, A. A. Few, G. A. Morris, W. J. Trabucco, and E. W. Paschal, Ground based instrumentation for measurements of atmospheric conduction current and electric field at South Pole, *J. Geophys. Res.*, 98, 2611-2618, 1993.
4. Deshpande, C. G., and A. K. Kamra, Diurnal variations of the atmospheric electric field and conductivity at Maitri, Antarctica, *J. Geophys. Res.*, 106, 14,207-14,218, 2001.
5. Israel, H., Atmospheric Electricity, vol. II, Isr. Program for Sci. Transl., Jerusalem, 1973.
6. Kasemir, H. W., Measurement of air-earth current density, in Proceedings of the conference on atmospheric electricity Geophysics Research, Paper No 42, Air-Force Research Center, Cambridge, Bedford, Mass 1955.
7. Panneerselvam, C., K. U. Nair, K. Jeeva, C. Selvaraj, S. Gurubaran, and R. Rajaram., A comparative study of atmospheric Maxwell current and electric field from a low latitude station, Tirunelveli, *Earth Planets Space*, Vol. 55 (No. 11), pp. 697-703, 2003.
8. Reiter, R., Fields, Currents and Aerosols in the Lower Troposphere, published for the division of Atmospheric Sciences, National Science Foundation, Washington, D.C., by Amerind Publishing Co. Pvt. Ltd., New Delhi 1985.
9. Ruhnke, L., H., Area averaging of atmospheric electric currents, *J. Geomagn. Geoelectr.*, 21, 453-462, 1969.
10. Willett, J.C., and J.C Bailey, Contact-potential and surface charge effects in atmospheric-electrical instrumentation, NRL, Memo. Rep. 5063, Nov. Res. Lab., Washington, D.C., April 1983.
11. Wilson, C. T. R., The electric field of the thundercloud and some of its effects, *Proc. Phys. Soc. London, Sect. A*, 111, 32D, 1925.